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J. Chem. Phys. 108, 6580–6587 (1998)

<https://doi.org/10.1063/1.476072>



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Improved Feynman's path integral method with a large time step: Formalism and applications

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(Received 1 December 1997; accepted 23 January 1998)

We describe an efficient path integral scheme for calculating the propagator of an arbitrary quantum system, as well as that of a stochastic system in special cases where the Fokker–Planck equation obeys strict detailed balance. The basic idea is to split the respective Hamiltonian into two exactly solvable parts and then to employ a symmetric decomposition of the time evolution operator, which is exact up to a high order in the time step. The resulting single step propagator allows rather large time steps in a path integral and leads to convergence with fewer time slices. Because it involves no system-specific reference system, the algorithm is amenable to all known numerical schemes available for evaluating quantum path integrals. In this way one obtains a highly accurate method, which is simultaneously fast, stable, and computationally simple. Numerical applications to the real time quantum dynamics in a double well and to the stochastic dynamics of a bistable system coupled to a harmonic mode show our method to be superior over the approach developed by the Makri group in their quasiadiabatic propagator representation, to say nothing about the propagation scheme based on the standard Trotter splitting. © 1998 American Institute of Physics.

[S0021-9606(98)02516-1]

I. INTRODUCTION

Solution of multidimensional problems of quantum and statistical mechanics remains a major challenge for theoreticians. Various different basis set methods and wave function propagation schemes have proven to be extremely valuable for the frontal attack on the Schrödinger and Fokker–Planck equations. However, the numerical work in these techniques scales unfavorably with the dimensionality of the system. The use of a short time propagator^{1–4} in conjunction with Monte Carlo integration⁵ is, at least in principle, the optimal approach. The advantage of this method is that it avoids storing large-dimensional Hamiltonian matrices and wave function vectors and thus allows one to treat multidimensional systems without introducing uncontrolled approximations.

The starting point for the discrete path integral approach is the observation that the propagator for a net increment t can be obtained by multiplying together N propagators each for the shorter time increment $\tau = t/N$. The latter can then be well approximated by a variety of methods (for recent reviews, see Refs. 2 and 4) to yield in a coordinate representation a multidimensional integral reading,

$$\psi(\mathbf{x}, t) = \int \prod_{m=1}^N d\mathbf{x}^{m-1} \psi_k(\mathbf{x}^m, \tau | \mathbf{x}^{m-1}) \psi(\mathbf{x}^0, 0) + O(t^{k+1}/N^k), \quad (1.1)$$

with $\mathbf{x}^N = \mathbf{x} \equiv (x_1, \dots, x_n)$. Here k denotes the order of approximation taken for the single step propagator,

$$\psi(\mathbf{x}^m, \tau | \mathbf{x}^{m-1}) = \psi_k(\mathbf{x}^m, \tau | \mathbf{x}^{m-1}) + O(\tau^{k+1}), \quad (1.2)$$

and $\psi(\mathbf{x}, t | \mathbf{x}^0)$ is the matrix element of the time evolution operator, $\psi(\mathbf{x}, t | \mathbf{x}^0) = \langle \mathbf{x} | e^{-\varepsilon H} | \mathbf{x}^0 \rangle$, where ε is imaginary ($\varepsilon = it$) or positive ($\varepsilon = t$) depending on whether quantum or statistical mechanics is concerned. Without loss of generality, we set in our subsequent consideration $\varepsilon = t$. Then, the matrix element of the quantum time evolution operator, $\langle \mathbf{x} | e^{-itH} | \mathbf{x}^0 \rangle$, follows from the present treatment by replacing $t \rightarrow it$. Besides, with the usual transcription $t \rightarrow \beta$, we are also considering the path integral representation of the Boltzmann operator $e^{-\beta H}$. Finally, the Boltzmann operator is often combined with the quantum time evolution operator in the calculation of time-dependent observables at finite temperature. This results in the complex time evolution operator $e^{-(it + \beta/2)H}$, whose path integral representation can be obtained from Eq. (1.1) by the substitution $t \rightarrow it + \beta/2$.

The practical applicability of numerical schemes for evaluation of the discrete path integral depends critically on the dimension of the latter. For this reason, short time propagators accurate for as long a time τ as possible are generally required to keep the number of integration variables in Eq. (1.1) as small as possible. The most commonly used short time propagator is obtained by partitioning the Hamiltonian operator,

$$H = -\frac{1}{2} D_{ii} \partial_{ii}^2 + V(\mathbf{x}), \quad (1.3)$$

into kinetic and potential energy terms,

$$H = T + V, \quad T = -\frac{1}{2} D_{ii} \partial_{ii}^2, \quad V = V(\mathbf{x}), \quad (1.4)$$

and employing the Trotter breakup to split the time evolution operator into a product of exactly solvable parts,

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$$e^{-\tau(T+V)} = e^{-\tau V/2} e^{-\tau T} e^{-\tau V/2} + O(\tau^3). \quad (1.5)$$

In the above, D_{ij} is a diagonal matrix with the elements $D_{ii} = \hbar^2/\mu_i$. Moreover, the summation rule over repeated indices is always implied. We will refer to Eq. (1.5) as the standard Trotter splitting. Of course, this splitting is not unique. But before discussing alternatives, it is worth noting the advantages and drawbacks of Eq. (1.5). The most appealing feature of the standard Trotter splitting is perhaps that it requires no analytical work to evaluate the short time propagator and can be applied to any Hamiltonian of the type (1.3), regardless of the particular form of the potential term. The use of Eq. (1.5) is equivalent to the conventional trapezoid rule discretization of the action in Feynman's path integral expression,⁶ and leads to a second-order coordinate propagator of the form

$$\begin{aligned} \psi_2(\mathbf{x}^m, \tau | \mathbf{x}^{m-1}) &= \psi_d(\mathbf{x}^m, \tau | \mathbf{x}^{m-1}) \\ &\times \exp\left\{-\frac{1}{2}\tau[V(\mathbf{x}^m) + V(\mathbf{x}^{m-1})]\right\}. \end{aligned} \quad (1.6)$$

Here ψ_d is the free-diffusion propagator

$$\begin{aligned} \psi_d(\mathbf{x}^m, \tau | \mathbf{x}^{m-1}) &= [(2\pi\tau)^n \det \mathbf{D}]^{-1/2} \\ &\times \exp\left[-\frac{1}{2\tau}(\mathbf{x}^m - \mathbf{x}^{m-1}) \right. \\ &\left. \cdot \mathbf{D}^{-1}(\mathbf{x}^m - \mathbf{x}^{m-1})\right]. \end{aligned} \quad (1.7)$$

It is also important to note that the above decomposition is in terms of unitary operators, so that the norm of ψ is preserved as the dynamics unfolds. This property makes a numerical algorithm based on the Trotter product formula very stable. Yet another advantage of Eq. (1.5) is that the resulting discrete path integral is well suited to fast Fourier Transform (FFT)⁷ and Monte Carlo integration schemes,^{1,2,5} though application of the latter to quantum dynamics is far from straightforward. The convergence problem, known as the "sign problem," is inherent in real time quantum dynamics even in the strict semiclassical limit. It lies in the oscillatory nature of the quantum propagator, or, in other words, in the lack of a good weight function from which to sample. A possible way of dealing with this problem is to introduce a sampling function based on some physical knowledge of the system, e.g., a Gaussian distribution around the classical path.² The disadvantage of the standard Trotter splitting is that the neglect of the commutator of T and V often introduces a large error that has to be compensated for by a small time step τ , and consequently a large number N of steps in the path integral. This limits considerably the time scale of many-body path integral calculations, reducing it to two or so periods of motion of quantum particles in dissipative environments.³ Another related limitation of the standard Trotter splitting is the necessity of doing the computations precisely, even if high precision is not demanded nor warranted. This is because the error made by using Eq. (1.6) rapidly grows with increasing τ , and beyond some $\tau_{\max}$, which is usually smaller than unity, the standard short time propagator fails to provide correct results.

An obvious way to refine Eq. (1.5) is to use an improved zeroth order representation, which would allow larger time steps in the path integral. For example, a number of methods have been developed to map a physical system described by a general potential energy function $V(\mathbf{x})$ onto a harmonic oscillator reference system, and to use its frequency as a variational parameter.^{6,8} The variational path integral approach is straightforward and can be applied to any Fokker-Planck and Schrödinger equations, regardless of the particular form of their coefficients. One drawback, however, is that it is often unclear how to judge the choice of the free parameter, or, in other words, the accuracy of the zeroth order representation, and then how to systematically improve upon the path integral solution without substantial computational effort. There has also been some recent success in path integral calculations with physically motivated anharmonic reference systems.^{3,9,10} The basic idea of this approach is the construction of effective system-specific propagators that are accurate for a particular problem, but much better behaved than the standard short time propagator. One can mention specifically the work of Makri and co-workers,^{3,10} who developed a path integral scheme for calculating dynamical properties of a one-dimensional system coupled to a harmonic bath. The starting point of their method is an improved quadiabatic propagator obtained numerically in terms of the Trotter product formula and a basis set method. Although system-specific propagators by construction incorporate the exact dynamics of physically motivated reference systems, this will not always assure us fast convergence (e.g., what if the system does not behave adiabatically at all?). The efficiency of this approach depends crucially on the decision about which part of the Hamiltonian properly describes the dynamics in the intermediate time domain. In some cases, a clear disparity between variables makes the choice obvious. For others, choosing the best partitioning of the system is more of an art than an exact science. Still, an improper choice can result in a less than optimally efficient propagator, or even one which is worse than the standard short time propagator.

Aside from using improved zeroth order representations, there are many other methods for obtaining better short time propagators, which also have the effect of reducing the number of integration variables. We mention specifically higher-order Trotter-like factorizations,¹¹ power series expansion techniques,^{12,13} extrapolation methods for removing time slice errors,^{14,15} and cumulant expansions.¹⁶ The key idea of the aforementioned methods is the construction of high-accuracy approximations for the propagator, correct to an order in $1/N$ as high as possible. An extensive study of their relative efficacy can be found in a previous paper.⁴ Here we only note that some of these methods are applicable to certain favorable situations, and all of them are more difficult to implement than the standard Trotter splitting.

The aim of the present paper is to show a simple alternative to the methods discussed above. The basic idea is to employ, instead of the Trotter breakup (1.5), a fourth-order symmetric decomposition of the time evolution operator designed for solving many-body problems of classical mechanics. The resulting short time propagator is particularly attrac-

tive for two reasons. First, it considerably improves the standard Trotter splitting, while retaining its principal advantages. Second, being used in a path integral, it makes possible quite large time steps with a mild loss of precision. The method is outlined in Sec. II. Two numerical examples demonstrating the power of the new technique are presented in Sec. III. Section IV concludes with general remarks and future applications.

II. FORMALISM

For the sake of generality we consider the Hamiltonian dynamics governed by

$$H = -\frac{1}{2}D_{ij}\partial_{ij}^2 + V(\mathbf{x}), \quad (2.1)$$

where D_{ij} is an arbitrary but positive definite diffusion matrix. With the usual transformation

$$P(\mathbf{x}, t | \mathbf{x}^0) = \sqrt{P_e(\mathbf{x})/P_e(\mathbf{x}^0)} \psi(\mathbf{x}, t | \mathbf{x}^0), \quad (2.2)$$

we are also considering the stochastic dynamics, $P(\mathbf{x}, t | \mathbf{x}^0) = \langle \mathbf{x} | e^{tL} | \mathbf{x}^0 \rangle$, governed by the Fokker–Planck operator

$$L = -\partial_i G_i(\mathbf{x}) + \frac{1}{2}D_{ij}\partial_{ij}^2, \quad (2.3)$$

provided that the latter obeys strict detailed balance¹⁷

$$L(P_e f) = P_e L^+ f \quad (2.4)$$

for any smooth function f . Here L^+ denotes the backward operator

$$L^+ = G_i(\mathbf{x})\partial_i + \frac{1}{2}D_{ij}\partial_{ij}^2, \quad (2.5)$$

while P_e is the equilibrium distribution satisfying $LP_e = 0$, which reads

$$P_e(\mathbf{x}) = Z^{-1} \exp \left[2 \int^{\mathbf{x}} d\mathbf{q} \cdot \mathbf{D}^{-1} \mathbf{G}(\mathbf{q}) \right], \quad (2.6)$$

with a normalization constant Z . If the requirement (2.4) is met, the Fokker–Planck operator (2.3) can be cast into the Hermitian form (2.1), where the potential $V(\mathbf{x})$ is determined in terms of the drift coefficients G_i and the inverse diffusion matrix D^{ij} by

$$V(\mathbf{x}) = \frac{1}{2} [D^{ij} G_i(\mathbf{x}) G_j(\mathbf{x}) + \partial_i G_i(\mathbf{x})]. \quad (2.7)$$

As noted in the Introduction, Trotter-approximated propagators by construction neglect the commutator of T and V . This neglect, however, is often quite severe and in general requires very fine discretization of the path integral. An obvious way for resolving this problem is to employ a higher-order factorization of the time evolution operator, which includes the commutator explicitly. Although factorizations of this kind have been known since the early days^{18–20} and have been frequently used for solving many-body problems in classical mechanics,²¹ their effective application to quantum and statistical mechanics was rare up to recent times.^{19,22}

To be specific, we choose a fourth-order symmetric decomposition of the form²¹

$$e^{\tau(A+B)} = e^{\tau A/6} e^{\tau B/2} e^{\tau C} e^{\tau B/2} e^{\tau A/6} + O(\tau^5),$$

$$C = \frac{2}{3}A + \frac{1}{72}\tau^2[A, [B, A]], \quad (2.8)$$

where A and B are two arbitrary noncommuting operators assumed to be bounded on a finite-dimensional Hilbert space. It may be noted that this factorization has already been used by the author for solving Fokker–Planck equations with singular diffusion matrices.²² In such a case, the Fokker–Planck operator cannot in principle be cast into a Hermitian form, and the utility of Eq. (2.9) must be studied for each particular system separately. By contrast, its application to Hamiltonian systems is straightforward and does not require any analytical work to evaluate the propagator. Indeed, using the same partitioning of the Hamiltonian as in Eq. (1.4), i.e., $A = V = V(\mathbf{x})$ and $B = T = -\frac{1}{2}D_{ij}\partial_{ij}^2$, one obtains

$$e^{-\tau H} = e^{-\tau V/6} e^{-\tau T/2} e^{-\tau \tilde{V}} e^{-\tau T/2} e^{-\tau V/6} + O(\tau^5), \quad (2.9)$$

where the function $\tilde{V}(\mathbf{x})$ can always be evaluated analytically in terms of the potential term $V(\mathbf{x})$ and its first derivatives:

$$\tilde{V} = \frac{2}{3}V + \frac{1}{72}\tau^2 D_{ij}(\partial_i V)(\partial_j V). \quad (2.10)$$

As one can see, the implementation of Eqs. (2.9) and (2.10) does not yield a more complicated result than that of the standard Trotter splitting, Eq. (1.5). In the coordinate representation the decomposition takes the form

$$\begin{aligned} \psi_d(\mathbf{x}^m, \tau | \mathbf{x}^{m-1}) \\ = \exp \left\{ -\frac{1}{6}\tau [V(\mathbf{x}^m) + V(\mathbf{x}^{m-1})] \right\} \int d\mathbf{x} \psi_d(\mathbf{x}^m, \frac{1}{2}\tau | \mathbf{x}) \\ \times \exp[-\tau \tilde{V}(\mathbf{x})] \psi_d(\mathbf{x}, \frac{1}{2}\tau | \mathbf{x}^{m-1}), \end{aligned} \quad (2.11)$$

with ψ_d defined by Eq. (1.7). Use of Eq. (2.11) in a path integral will require two FFT evaluations (Monte Carlo sampling) of Gaussian variables rather than one, as is the case for Eq. (1.6). But the resulting error is of order $1/N^4$; consequently, the present factorization should allow larger time steps to be taken than the Trotter breakup for comparable accuracy. Thus the key question we shall address in our numerical applications is: Does the increase in time step more than compensate for the added computational complexity?

To conclude this section it should be noted that the above decomposition is symmetric. Therefore, extrapolation methods can be used to further improve its accuracy.^{14,15,22} Besides, various different (harmonic or anharmonic) representations of the propagator can also be constructed in terms of Eq. (2.9) by splitting the Hamiltonian into a leading (reference) contribution H_0 and a correction $H_1 = H - H_0$. However, we will not do so here to keep the presentation simple. Our purpose is to show that, even with the standard splitting, the present treatment produces propagators much more efficient than the best of the Trotter-approximated propagators now in use.

III. NUMERICAL RESULTS

The aim of this section is twofold: First, we want to test the convergence of our path integral method. And second, we wish to compare its efficacy with that of other path integral techniques based on the Trotter product formula. To this end, we consider two problems, namely, the real time quan-

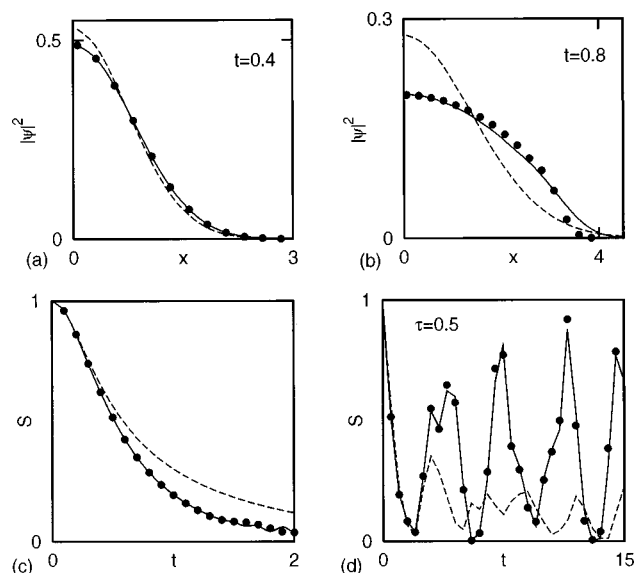


FIG. 1. The modulus squared of the evolved wave function for positive-valued x , $|\psi(x,t)|^2 = |\psi(-x,t)|^2$, at two propagation times ($t=0.4$ and 0.8), and the survival probability, $S(t)$ [Eq. (3.3)], for the symmetric double well potential, Eq. (3.1), obtained by using the standard Trotter splitting (dashed lines) and the present fourth-order approximation (solid lines) as single step propagators [plots (a), (b), and (c)]. Circles are for exact results generated by basis set expansion. Plot (d) shows the survival probability obtained in the path integral evaluation with the same propagators for $\tau=0.5$.

tum dynamics of a one-dimensional bistable system and the stochastic dynamics of a double well coupled to a harmonic mode. The examples are taken simple enough in order to allow comparison with a standard basis set method without extraordinary computational effort.

A. Real time propagator

To begin, we consider the real time quantum dynamics in a one-dimensional double well potential of the form

$$V(x) = \frac{1}{4}x^4 - 2x^2. \quad (3.1)$$

In our calculation we use dimensionless units with $\hbar = \mu = 1$. The initial wave function is taken to be a Gaussian:

$$\psi(x,0) = (\alpha/\pi)^{1/4} \exp(-\frac{1}{2}\alpha x^2), \quad (3.2)$$

with $\alpha=5$. The implementation of Eq. (2.11) for the real time propagator $\langle x|e^{-itH}|x_0\rangle$ is straightforward. To this end, it is enough to replace in the right-hand side of Eq. (2.11) t with it . The quadratures called for in the evaluation of the path integral, Eq. (1.1) with (2.11), are done by using the FFT.

Figure 1 shows the time evolution of the square of the wave function, $|\psi(x,t)|^2$, and compares it with exact results, generated by a basis set method. The calculation is performed using Eqs. (1.6) and (2.11) as *single step propagators*. We depict only the range $x \geq 0$, since $\psi(x,t) = \psi(-x,t)$ in the considered case. For a more quantitative test of the quality of the evolved wave function we also show in the figure the survival probability $S(t)$ for the time evolution of a state initially described by the wave function $\psi(x,0)$:

$$S(t) = \int_{-\infty}^{\infty} dx |\psi(x,t)\psi(x,0)|^2. \quad (3.3)$$

The excellent agreement demonstrates the potential of the proposed method. It is seen to be capable of describing correctly the quantum dynamics of a very anharmonic process over a broad range of t . However, the primary purpose of Eq. (2.11) is an improved short time propagator for the use in a path integral. Figure 1(d) shows the time evolution of the survival probability obtained in the path integral evaluation according to Eq. (1.1). As evidenced by the figure, the excellent agreement at all times is achieved with the present treatment even though the time step is rather large, $\tau=0.5$. In contrast, the standard Trotter splitting fails completely in this case. An acceptable level of accuracy is attainable in calculations with Eq. (1.6) only for $\tau \leq 0.1$. Consequently, the standard short time propagator involves rather long computation if the desired propagation time is large. The use of Eq. (2.11), however, allows the time evolution to be followed for longer time with the same number of time slices. We believe that the higher-order propagator described in this paper will substantially extend the time scale of quantum path integral calculations and thus the variety of problems that are computationally accessible.

B. System-bath problem

As a second and more challenging example, we consider a two-dimensional stochastic model that has been put forth for charge-transfer reactions in polar solvent.²³ There the influence of the nonequilibrium solvation on the reaction dynamics can be properly described by an additional solvation coordinate y that couples to the reactive coordinate x through the potential of mean force

$$\Phi(x,y) = U(x) + \frac{1}{2}\gamma(x-y)^2. \quad (3.4)$$

Here $U(x)$ is a bistable potential, given by

$$U(x) = \frac{1}{4}x^4 - \frac{1}{2}x^2, \quad (3.5)$$

and γ the coupling constant. When the motion of the remaining degrees of freedom that make up the heat bath is fast, the dynamics can be modeled in terms of a Fokker–Planck equation whose operator in dimensionless variables reads

$$L = \partial_x[U'(x) + \gamma(x-y) + \frac{1}{2}D\partial_x] + \partial_y[\gamma(y-x) + \frac{1}{2}D\partial_y]. \quad (3.6)$$

In the last expression, the prime denotes the derivative with respect to x and D is a dimensionless temperature. In deriving Eq. (3.6) we have assumed, for simplicity, that the damping coefficients in the direction of system and bath coordinates are equal to each other. It may also be noted that the potential $\Phi(x,y)$ defined by Eqs. (3.4) and (3.5) has two minima at $(-1,-1)$ and $(1,1)$ and a transition point at the origin.

The explicit expression for the equilibrium distribution is

$$P_e(x,y) = Z^{-1} \exp[-2\Phi(x,y)/D]. \quad (3.7)$$

With the ansatz (2.2), the Fokker–Planck operator (3.6) is transformed into a Hermitian form, which reads

$$H = -\frac{1}{2}D\partial_{xx}^2 + \frac{1}{4}D^{-1}U'^2(x) - \frac{1}{2}U''(x) - \frac{1}{2}D\partial_{yy}^2 + \gamma^2 D^{-1}[y-x-U'(x)/(2\gamma)]^2 - \gamma. \quad (3.8)$$

In a previous paper,²⁴ the above model was studied numerically over a large range of parameters by means of a basis set method. The quantity of interest is the probability density of finding the system at time moment t at position x ,

$$P(x, t) = \int_{-\infty}^{\infty} dy P(x, y, t). \quad (3.9)$$

We found that the dynamics described by the model crucially depends on the relative order of the parameters D and γ . If, for example, the diffusion constant D is small, $D \ll 1$, while the coupling is strong, $\gamma \geq 2$, the smallest nonvanishing eigenvalue is well separated from the others and given by the Kramers–Langer expression for the transition rate between the two metastable states of the potential $\Phi(x, y)$.²⁵ It consequently is of Arrhenius type, with an activation energy given by the barrier height at the saddle point, $E = (2D)^{-1}$. If, however, the coupling γ is not large, $\gamma \leq 1$, a slow one-dimensional motion in the y -direction in an effective double well potential results, which has a smaller barrier height than the bare potential. The longtime behavior is then also governed by a single Arrhenius type eigenvalue but with a smaller activation energy than that following from the bare potential. Finally, in the limit of weak coupling, $\gamma \leq D$, the dynamics is governed by a densely spaced spectrum of almost equidistant eigenvalues. The longtime behavior is generally multiexponential (barrierless) in this case even though the bare potential height E might be large.²⁴

It is our aim here to calculate the time evolution of $P(x, t)$ using different approximations for the short time propagator. To be specific, we set $x_0 = y_0 = 1$, $D = 0.1$, and performed the calculation in all regimes of interest, reaching from deep intrawell relaxation (slow harmonic mode) to interwell thermal activation (fast harmonic mode). But before presenting our results a few remarks on the numerical procedures used are in order.

First, we note that the Hamiltonian operator given by Eq. (3.8) is of system-bath form. Consequently, the quasiadiabatic representation of the propagator, suggested by Makri and co-workers,^{3,10} can be used in this case. It is obtained by partitioning the Hamiltonian into a system part,

$$H_x = -\frac{1}{2}D\partial_{xx}^2 + \frac{1}{4}D^{-1}U'^2(x) - \frac{1}{2}U''(x), \quad (3.10)$$

and the rest

$$H_y = -\frac{1}{2}D\partial_{yy}^2 + \gamma^2 D^{-1}[y-x-U'(x)/(2\gamma)]^2 - \gamma, \quad (3.11)$$

and employing the Trotter product formula

$$\exp[-\tau(H_x + H_y)] = \exp(-\frac{1}{2}\tau H_y) \exp(-\tau H_x) \times \exp(-\frac{1}{2}\tau H_y) + O(\tau^3). \quad (3.12)$$

This yields in a straightforward way

$$\begin{aligned} \psi_2(x_m, y_m, \tau | x_{m-1}, y_{m-1}) \\ = \langle x_m | e^{-\tau H_x} | x_{m-1} \rangle \sqrt{\frac{\omega e^{2\gamma\tau}}{2\pi D \sinh \omega \tau}} \exp \frac{\omega}{2D \sinh \omega \tau} \\ \times \{2(y_m - s_m)(y_{m-1} - s_{m-1}) - (s_m - s_{m-1}) \\ \times \cosh^2 \frac{1}{2}\omega\tau - 2(s_m - s_{m-1})(y_m - y_{m-1} + s_{m-1} - s_m) \\ \times \cosh \frac{1}{2}\omega\tau - [(y_m - s_m)^2 + (y_{m-1} - s_{m-1})^2] \\ \times \cosh \omega\tau\}, \end{aligned} \quad (3.13)$$

where $s_m = x_m + U'(x_m)/(2\gamma)$ and $\omega = \gamma\sqrt{2}$. The propagator for the reference system H_x that appears in the last equation is constructed numerically on a grid of points $\{x_m\}$ by a basis set method. It is clear that the above quasiadiabatic approximation is much more difficult to implement than Eq. (1.6). The resulting discrete path integral, Eq. (1.1) with Eq. (3.13), is suited neither to FFT nor to Monte Carlo techniques, and must be calculated by propagator matrix multiplication. Moreover, in common with the standard short time propagator, this quasiadiabatic approximation is based on the primitive Trotter breakup and therefore neglects the commutator of T and V . One may thus expect that, excepting some favorable regime of parameter space, it will work no better than Eq. (1.6).

Next, it may be noted that within the present treatment the integral over the bath mode in Eq. (2.11) is Gaussian. Therefore, it can be integrated out analytically, leaving us with an integral over the system coordinate only. However, we will not do so here. Instead, we evaluate Eqs. (1.1) and (2.11) numerically by taking advantage of the FFT. The same technique is also used for numerically treating the path integral based on the standard short time propagator. The favorable scaling of the FFT, which is almost linear with the total number of grid points, makes this approach much more efficient than propagator matrix multiplication techniques.

Finally, we would like to emphasize that the decompositions used above to factorize the time evolution operator by construction preserve the norm of the auxiliary function ψ rather than that of the distribution function P itself. The normalization condition for the latter reads

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dx dy P(x, y, t) = 1. \quad (3.14)$$

In our calculations we enforced Eq. (3.14) by a simple procedure of renormalizing the distribution function so that the norm is preserved.¹⁶

Figure 2 shows the time evolution of $P(x, t)$, when employing different P_k as *single step propagators*, and compares it with exact results generated by basis set expansion for $\gamma = 0.1, 1$, and 3 . As anticipated, for a weak coupling ($\gamma = 0.1$) the quasiadiabatic numerical propagator (3.13) is more accurate than the standard Trotter splitting (1.6). With increasing γ , however, the error made by Eq. (3.13) also increases and for $\gamma \geq 1$ the quasiadiabatic representation becomes as bad as the standard Trotter splitting. This is in drastic contrast to the present fourth-order propagator, Eq. (2.11). The latter remains accurate over a larger range of t

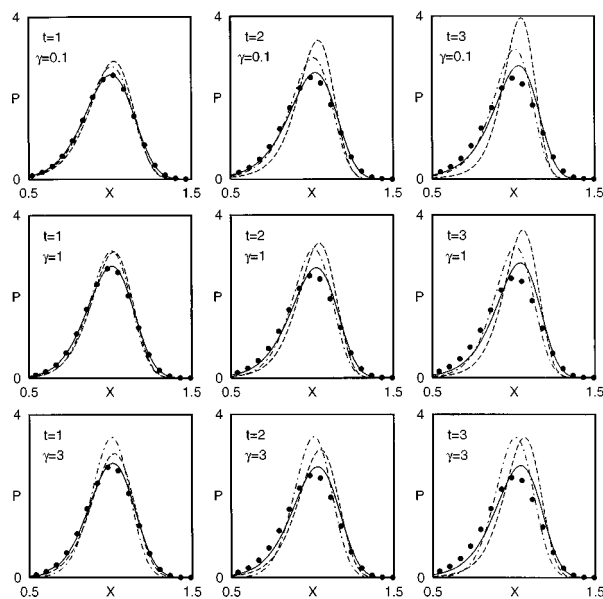


FIG. 2. Probability distributions [as defined in Eq. (3.9)] obtained in terms of single step propagators for $t=1, 2$, and 3 and $\gamma=0.1, 1$, and 3 . Circles: exact results; dashed lines: standard Trotter splitting, Eq. (1.6); dot-dashed lines: quasiadiabatic propagator, Eq. (3.13); solid lines: present fourth-order approximation, Eq. (2.11).

than both Trotter-approximated propagators and provides the best agreement with the exact numerical results whatever the coupling constant is.

The above observations are also well illustrated by Fig. 3, which shows the relative error for the second cumulant defined by

$$\langle\langle x(t) \rangle\rangle = \langle x^2(t) \rangle - \langle x(t) \rangle^2, \quad (3.15)$$

where

$$\langle x^n(t) \rangle = \int_{-\infty}^{\infty} dx x^n P(x, t). \quad (3.16)$$

This quantity, characterizing the width of the distribution function, is more sensitive to the error made in Eq. (3.16) by using an approximate single step propagator than the moments $\langle x(t) \rangle$ and $\langle x^2(t) \rangle$ themselves. As evidenced by the figure, the most accurate results are obtained with the present fourth-order approximation. It is seen to be superior over the second-order quasiadiabatic propagator, Eq. (3.13), even though $\gamma \ll 1$. With increasing coupling constant, the accuracy of Eq. (2.11) deteriorates rather slowly, and for $\gamma=3$ it remains still accurate over a large interval of t . By contrast, the error made by Eq. (3.13) increases with γ very rapidly, making this approximation of no practical importance already for $\gamma \gtrsim 2$.

Next, we apply the various approximations discussed above for the single step propagator to the path integral evaluation of the same quantity. Shown in Fig. 4 are results obtained for $\gamma=0.1, 1$, and 2 , with time discretizations as coarse as $\tau=0.5$ and 1 . As expected, the quasiadiabatic propagator works well only in the weak coupling limit $\gamma \rightarrow 0$. For example, for $\tau=0.5$ and $\gamma=0.1$, using the improved zeroth order representation in place of the standard

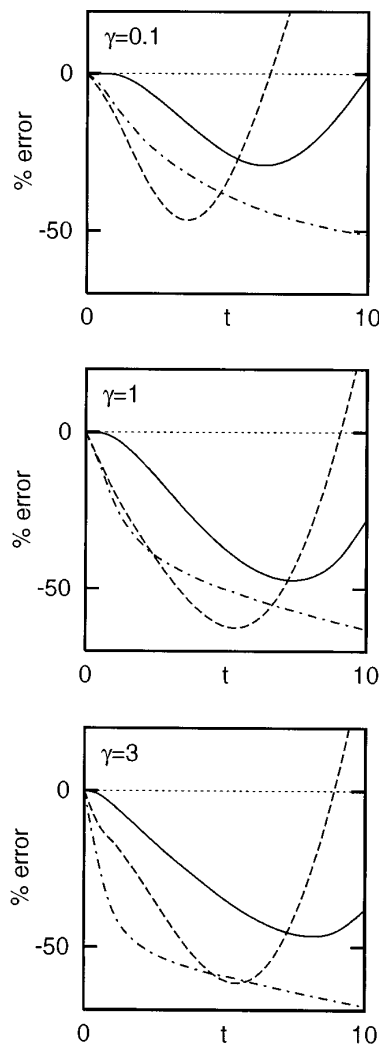


FIG. 3. Same as in Fig. 2, but for the percentage error, $100 \times (\text{approximate} - \text{exact}) / \text{exact}$, in the second cumulant [Eq. (3.15)].

splitting reduces the error by a factor of 4. Although no system-specific reference system is employed in the present treatment, the effect of accounting for the commutator of T and V is the further reduction of the error roughly by one order of magnitude. Thus the resulting error is of the order of 0.7%. It is pleasing that this precision is attained with a rather large time step $\tau=0.5$. For comparison, the standard short time propagator provides an analogous accuracy for $\tau \lesssim 0.08$ and thereby requires a computation that is three times as large. The same is true for the quasiadiabatic propagator, but in that case the necessary computational effort is much larger. This is because the use of Eq. (3.13) in the path integral requires a propagator matrix multiplication technique, which is considerably less efficient than the FFT.

It is also remarkable that, with increasing coupling constant, the error made by using the present propagator only slowly increases, reaching 10% for $\gamma=2$. The same is not true for the Trotter-approximated propagators, which are seen to be completely inadequate for $\tau=0.5$ and $\gamma=2$. In this case, an accuracy of 10^{-1} is attainable with Eqs. (1.6) and (3.13) for $\tau \lesssim 0.1$. Whereas with further increasing coupling constant, $\gamma \gtrsim 3$, both short time approximations fail to

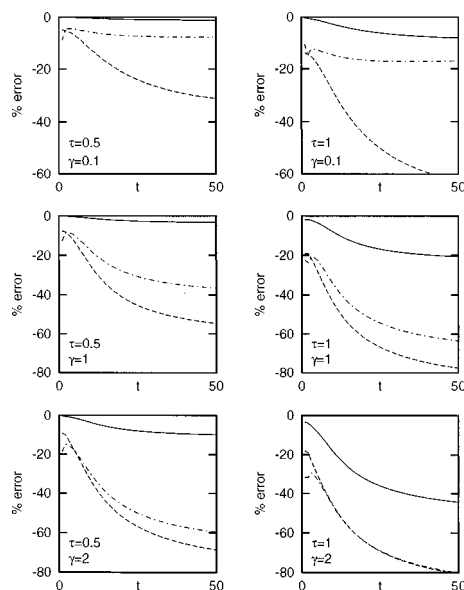


FIG. 4. Percentage error in the path integral evaluation of the second cumulant, Eq. (3.15). The notations are the same as in Fig. 2.

provide good resolution even though $\tau=0.1$. One may thus conclude that the system-specific representation of the propagator developed by Makri and co-workers,^{3,10} Eq. (3.13), preserves its significance just for vanishing coupling ($\gamma \rightarrow 0$). In that limit, the quasiadiabatic representation of the propagator indeed is accurate over a broader range of t than the standard Trotter splitting. Other than that, the quasiadiabatic propagator works no better (and even worse) than the standard one, Eq. (1.6).

Before closing we would like to emphasize that due to the numerical efficiency of the present technique, a relative error of 20% is attainable for $\gamma \lesssim 1$ with quite a large time step $\tau=1$. As evidenced by Fig. 4, the quasiadiabatic propagator also provides the same level of accuracy with $\tau=1$, but only for $\gamma \lesssim 0.1$. Finally, the error made by the standard Trotter splitting is seen to be very large for all γ 's in this case.

IV. CONCLUDING REMARKS

We presented a new path integral methodology for solving multidimensional Schrödinger and Fokker-Planck equations. It is based on a fourth-order symmetric decomposition of the time evolution operator designed for solving many-body problems of classical mechanics. We showed that the application of this decomposition to quantum and statistical dynamics is straightforward and does not yield a result more complicated than that of the standard Trotter splitting. The method is very accurate and easily applicable to a general class of Hamiltonian operators, Eq. (2.1), regardless of the particular form of the potential term. If time evolution is needed for only relatively short times, then it may be possible to use the respective higher-order propagator for the entire time increment. However, its primary purpose is an improved short time propagator for use in a path integral. Implementing it numerically requires no special algorithms and can be carried out very efficiently in terms of conventional FFT and Monte Carlo integration schemes. The power

of the method is illustrated by considering the quantum real time dynamics in a double well and the stochastic dynamics of a two-dimensional model describing charge-transfer reactions in polar solvents. The results obtained are very encouraging, since the dynamics of the models can be quite accurately described with rather large time steps in all regimes of physical interest. Our numerical comparison also includes results from two Trotter-approximated propagators based on different splittings of the Hamiltonian operator. The numerical applications clearly demonstrated the present approach to be superior in the sense that its utility is *not* restricted to a certain favorable regime of parameter space, while the method itself yields much more *accurate* results with *less* computational effort than the best alternative Trotter-approximated propagator now in use.

The formalism presented in this paper is indeed rigorous. But its strengths and limitations when treating complex many-body problems have yet to be explored. In such a case, an important point deserving consideration is whether the increase in time step compensates for the computational load in evaluating the correction term $[V, [T, V]] = D_{ij}(\partial_i V) \times (\partial_j V)$.

As noted, the efficacy of our method can be further enhanced by making use of improved zeroth order representations. For example, a fourth-order breakup with a harmonic-oscillator reference system is easily constructed by splitting the full Hamiltonian into a harmonic contribution $A = H_0$ and an anharmonic correction $B = V_1$ reading

$$H_0 = -\frac{1}{2}D_{ij}\partial_{ij}^2 + \frac{1}{2}\omega^2 D^{ij}x_i x_j,$$

$$V_1 = V(\mathbf{x}) - \frac{1}{2}\omega^2 D^{ij}x_i x_j.$$

The respective short time approximation takes the same form as in Eq. (2.11) but with the prescriptions $V \rightarrow V_1$ and $\psi_d \rightarrow \psi_0$, where ψ_0 is the reference harmonic-oscillator propagator, $\psi_0(\mathbf{x}, t | \mathbf{x}^0) = \langle \mathbf{x} | e^{-tH_0} | \mathbf{x}^0 \rangle$. Although quasiharmonic path integral representations are, in general, not suited to the FFT, a way for overcoming this problem has been developed in a recent paper.¹⁵ Its key idea is to employ the Stirling interpolation to dynamically readjust the distribution function every time step with a mild increase in cost and with no loss of precision. The affiliated frequency ω appearing in the above equations is a free parameter, which can be determined according to variational principles.^{6,8,13,15}

Quasiadiabatic representations like that based on Eqs. (3.10) and (3.11) can also be used within the present approach to efficiently treat the dissipative dynamics of system-bath Hamiltonians. In such a case, the evaluation of $[V, [T, V]]$ does not present any problem and we expect to obtain good results for a not too strong coupling between the system and the bath coordinates using the respective higher-order quasiadiabatic approximation as a *single step propagator*. Otherwise, a path integral propagation scheme can be employed to cover the whole parameter space. In the particular case of a *harmonic* bath, the discrete path integral can be evaluated rather efficiently using either Feynman's influence functional methodology^{6,3,10} or a flexible Monte Carlo scheme developed by Cao and co-workers.²⁶

Finally, we would like to emphasize that we have not explicitly covered all possible cases to which our approach would apply. Besides Hamiltonian systems, it can also be applied to Fokker–Planck equations with singular diffusion matrices, which are, in principle, not reducible to a Hermitian form. The latter are known to play an important role in the Kramers theory of chemical reactions²⁵ and its various generalizations to non-Markovian activated rate processes.²⁷ We plan to use our methodology to gain insight into a number of questions related to this theory.²⁸

ACKNOWLEDGMENT

We acknowledge the financial support of the Dirección General de Investigación Científica y Técnica of Spain.

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